

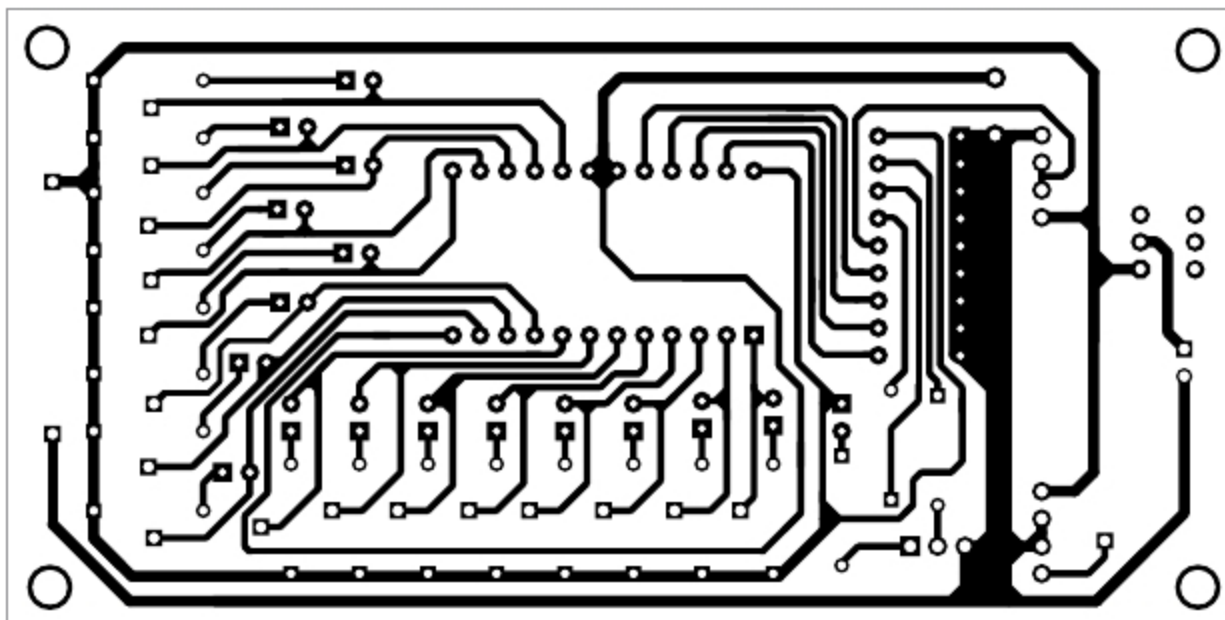


Fig. 6: Actual-size PCB layout of receiver circuit

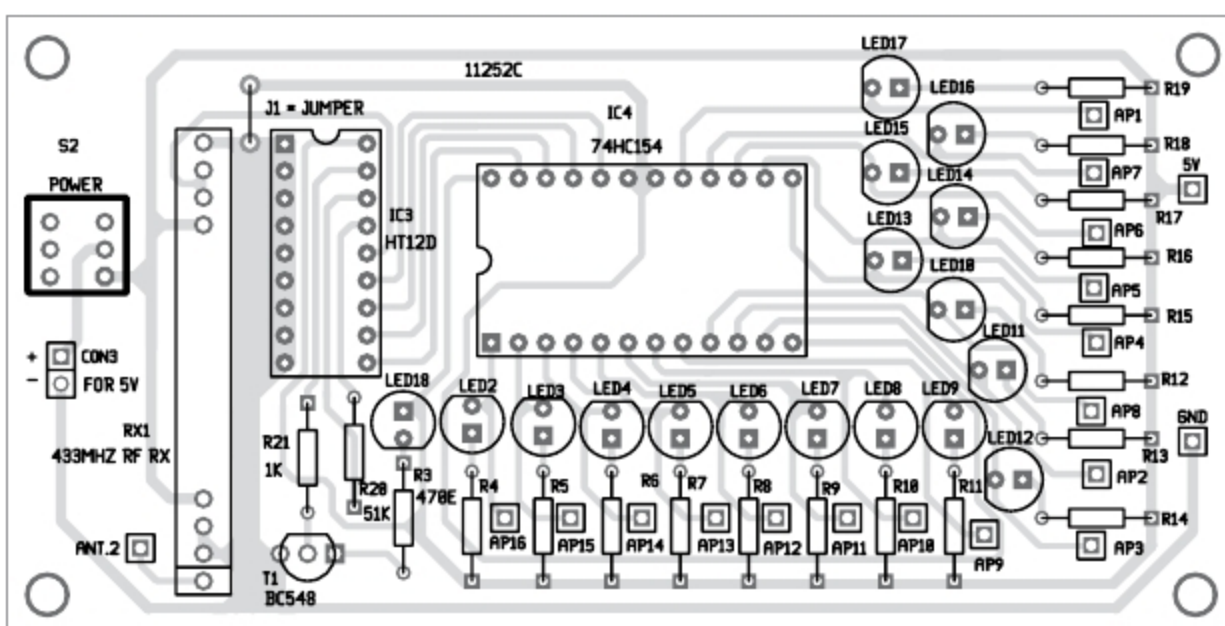


Fig. 7: Components layout for the PCB shown in Fig. 6

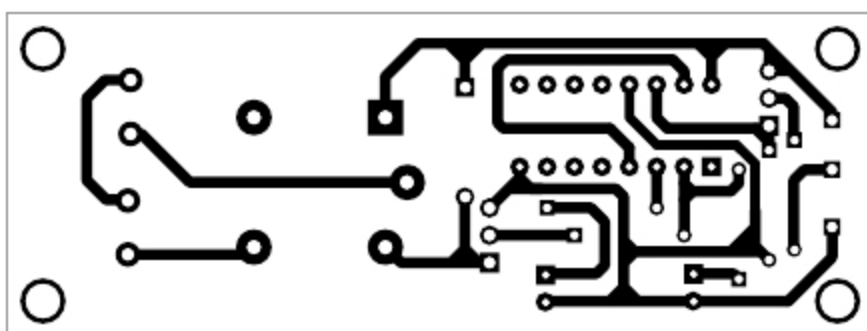


Fig. 8: Actual-size PCB layout of bistable circuit

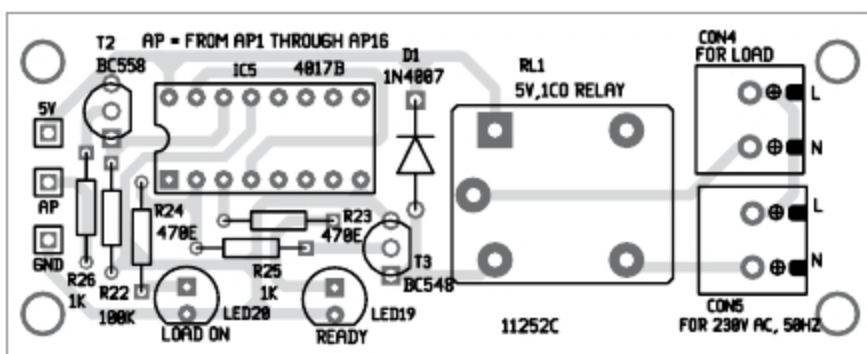


Fig. 9: Components layout for the PCB shown in Fig. 8

will go high, RL1 will be de-energised and load will be turned off and LED17 and LED20 will also be off.

LEDs connected in these circuits

the circuit on the designed PCB. Also connect RX1 to the PCB.

An actual-size PCB layout of the bistable circuit is shown in Fig. 8

have a particular purpose, as shown in the table.

Construction and testing

An actual-size PCB layout of the transmitter circuit is shown in Fig. 4 and its components layout in Fig. 5. Assemble the circuit on the designed PCB. Also connect TX1 to the PCB.

An actual-size PCB layout of the receiver circuit is shown in Fig. 6 and its components layout in Fig. 7. Assemble

PARTS LIST

Semiconductors:

IC1	- MM74C922 key encoder
IC2	- HT12E encoder
IC3	- HT12D decoder
IC4	- 4-to-16-line decoder 74HC154
IC5	- 4017B decade counter
D1	- 1N4007 rectifier diode
T1, T3	- BC548 npn transistor
T2	- BC558 pnp transistor
LED1-LED20	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 1-mega-ohm
R2, R4-R19	- 330-ohm
R3, R23, R24	- 470-ohm
R20	- 51-kilo-ohm
R21, R25, R26	- 1-kilo-ohm
R22	- 100-kilo-ohm

Capacitors:

C1, C2, C4	- 0.1 μ F ceramic disk
C3	- 10 μ F, 16V electrolytic

Miscellaneous:

CON1	- 8-pin terminal connector
CON2, CON3	- 2-pin connector
CON4, CON5	- 2-pin terminal connector
TX1	- 433MHz RF transmitter module
RX1	- 433MHz RF receiver module
ANT.1, ANT.2	- 13cm wire
S1, S2	- On/off switch
RL1	- 5V, single-changeover relay
	- 4x4 keypad
	- Female-to-female jumper wires
	- 20-pin Berg strip connector
	- 5V regulated power supply
	- Heatsink for 7805
	- 230V-operated appliance
	- 230V AC, 50Hz
	- 16-pin IC socket
	- 18-pin IC socket
	- 20-pin IC socket

and its components layout in Fig. 9. Assemble the circuit on the designed PCB. You can assemble 16 bistable units with the same circuit depending on your requirement for the number of appliances you would like to control.

After assembling all three units, connect the receiver unit with the bistable unit using an external connector. Connect these to each output of IC4. Also connect the keypad across CON1 on transmitter side—one antenna each on TX1 and RX1. Connect 230V AC, 50Hz across CON5, and load/appliance across CON4. Now, the circuit is ready to use. **EFY**



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